

SKILL CERTIFICATE

CAREER AND TECHNICAL EDUCATION

This is to certify that
Emil Gruenwald

has demonstrated competency in

Computer Programming II C+

+

05/03/2024

1896



Sydnee Dickson
Sydnee Dickson
State Superintendent

Thalea Longhurst
Thalea Longhurst
State Career and Technical Education Director

Career Skills Certificate Report



Exam Details

EMIL GRUENWALD

Computer Programming II C++

Certification date: 05/03/2024

This document certifies that the student listed has
acquired the following skills and knowledge.

Skills and Knowledge

Standards

- Students will demonstrate static (array), dynamic (vector, ArrayList, etc.) list structures, and strings.
- Students will properly use sequential files.
- Create user defined functions using top-down design and functional decomposition.
- Students will properly demonstrate object-oriented programming techniques.
- Students will properly demonstrate code comprehension and debugging techniques.
- Students will apply appropriate programming skills as an effective member of a team demonstrating the ability to collaborate with others (www.p21.org).
- Students will be aware of career opportunities in the Computer Programming/Software Engineering industry and of its history.